

2013 VJC H1 Chemistry Prelim Exam 8872/2
Suggested Answers

Section A

Answer **all** the questions in this section in the spaces provided.

1

	Compound X	$\xrightarrow{\text{heat}}$	Compound Y	$\xrightarrow{\text{heat}}$	Compound Z
% by mass	oxygen 13.38% metal M 86.62%		oxygen 9.33% metal M 90.67%		oxygen 7.17% metal M 92.83%

Compound **X** is a dark brown compound which consists of oxygen and metal **M** only.
 Successive heating of compound **X** transformed it into two other compounds.

- (a) Calculate the mass of oxygen that is combined with 1.00 g of metal **M** in each of these three compounds.

Compound X contains 13.38 g of oxygen for every 86.62 g of metal M; by division, every 1.00 g of metal M will combine with $\frac{13.38}{86.62}$ g of oxygen = 0.1545 g

[same method for compound Y and Z; no working required for full credit]

In compound **X**, 1.00 g metal **M** combines with • **0.1545 g** of oxygen.

In compound **Y**, 1.00 g metal **M** combines with • **0.1029 g** of oxygen.

In compound **Z**, 1.00 g metal **M** combines with • **0.07721 g** of oxygen.

[2]

- (b) The empirical formula of compound **X** is **MO₂**.
 By considering the ratios of the amount of oxygen present in the three compounds, determine the empirical formulae for compound **Y** and compound **Z**.

$$\frac{\text{mass of oxygen in compound Y}}{\text{mass of oxygen in compound X}} = \frac{0.1029}{0.1545} = \frac{2}{3}$$

We have $\frac{2}{3}$ as much oxygen in compound **Y** compared to compound **X**;
 therefore, the empirical formula of compound **Y** is **M₃O₄** (formula is **MO_{3/4}** but we multiply both factors by 4 to get rid of the fraction).

$$\frac{\text{mass of oxygen in compound Z}}{\text{mass of oxygen in compound X}} = \frac{0.07721}{0.1545} = \frac{1}{2}$$

We have $\frac{1}{2}$ as much oxygen in compound **Z** compared to compound **X**;
 therefore, the empirical formula of compound **Y** is **MO**.

Compound **Y** **M₃O₄**

Compound **Z** **MO**

[3]

- (c) Identify metal **M** by calculating its molar mass.

Let the molar mass of metal **M** be x g/mol and consider the empirical formula of compound **X**.

$$\frac{\text{no. of moles of metal M}}{\text{no. of moles of oxygen}}$$

$$= \frac{1.00/x}{0.1545/16.0} = \frac{1}{2}$$

$$\therefore x = 207.2$$

Metal **M** is **lead** with a molar mass of **207.2 g mol⁻¹**.

[2]

[Total: 7]

- 2 This question is about halogens and its compounds.

- (a) World War I is sometimes known as 'The Chemists' War'. Knowledge of Chemistry was applied towards developing high explosives and new methods such as the large scale use of poison gas.

The first successful use of chlorine as a poison gas was at Ypres, Belgium, on 22 April 1915. 170 tonnes (1 tonne = 1000 kg) of chlorine contained in 5730 cylinders was released forming a grey-green cloud which drifted across French troops. Chlorine can damage the eyes, nose, throat and lungs and is fatal at concentrations of 1000 ppm and above (1 ppm = 1 mg dm⁻³). Early counter-measures to chlorine included instructing troops to cover their mouths with gauze pads soaked in sodium hydrogen carbonate solution. Eventually, more effective counter-measures to chlorine were developed and thus other poison gases were introduced.

- (i) Calculate the maximum amount of chlorine gas that could be released from one of the cylinders that was used at Ypres on 22 April 1915.

$$\text{Amount of chlorine in 1 cylinder} = \frac{170 \times 10^3 \times 10^3}{71.0} / 5730 = 4.18 \times 10^2 \text{ mol}$$

- (ii) Determine the concentration of chlorine gas, in mol dm⁻³, in 1000 ppm of the gas.

$$1 \text{ mg dm}^{-3} = 1 \times 10^{-3} \text{ g dm}^{-3}$$

$$\text{Concentration of Cl}_2 \text{ in g dm}^{-3} = 1000 \times 1 \times 10^{-3} = 1.00 \text{ g dm}^{-3}$$

$$\text{Concentration of Cl}_2 \text{ in mol dm}^{-3} = \frac{1}{71.0} = 0.0141 \text{ mol dm}^{-3}$$

- (iii) In an accident, the chlorine gas from one such cylinder was released into a factory room of volume 25.0 m³.

Determine if the concentration of chlorine gas was fatal. Assume that the gas was released at room temperature and pressure.

$$1 \text{ m}^3 = 1 \times 10^3 \text{ dm}^3$$

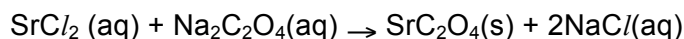
$$\text{Volume of factory room} = 25 \times 1 \times 10^3 \text{ dm}^3 = 25000 \text{ dm}^3$$

$$\text{Concentration of Cl}_2 \text{ in the factory} = \frac{4.18 \times 10^2}{25000} = 0.0167 \text{ mol dm}^{-3}$$

Since $0.0167 \text{ mol dm}^{-3} > 0.0141 \text{ mol dm}^{-3}$, it is fatal

[4]

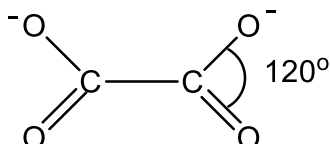
- (b) Strontium chloride is used to prepare strontium ethanedioate, SrC_2O_4 , which is widely used as a red colorant in fireworks. The equation for the reaction is as shown:



- (i) State the type of bonding present in SrC_2O_4 .

Ionic

- (ii) Deduce the shape and bond angle in the ethanedioate part of SrC_2O_4 by drawing its structural formula.



Trigonal planar wrt C and bond angle = 120°

- (iii) State the number of sigma and pi bonds present in the ethanedioate part of SrC_2O_4 .

5 sigma, 2 pi

[4]

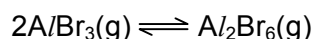
- (c) (i) Halogens can react with aluminium to form halides such as AlF_3 and AlBr_3 . When heated, aluminium bromide remains as a solid until 371K whereas aluminium fluoride remains as a solid until 1530K. Account for the difference in their melting points in terms of structure and bonding.

AlF_3 has a giant ionic lattice with strong ionic bonds between Al^{3+} and F^- ions. The strong ionic attractions require a lot of energy to break them apart hence the high melting point.

AlBr_3 has a simple covalent structure with the much weaker dispersion forces between the molecules hence very little energy is needed to melt the solid.

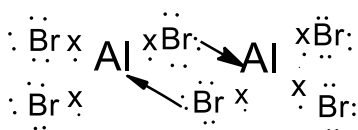
[3]

- (ii) In the vapour phase, an equilibrium is established between aluminium bromide and its dimer as follows:



With the help of a dot-and-cross diagram, explain how the dimer is formed.

The aluminium atom in AlBr_3 has only 6 electrons surrounding it hence it is electron-deficient. To complete the octet, the lone pair electrons from a bromine atom in a neighbouring molecule would be donated via a dative bond, resulting in a dimer.



[2]

[Total: 13]

3 This question is about Period 3 elements and their compounds.

(a) The ionic radii of four consecutive elements in period 3 are given below.

element	A	B	C	D
ionic radius / nm	0.065	0.050	0.041	0.212

- (i) Deduce which one of the elements, **A**, **B**, **C** or **D** corresponds to magnesium. Explain your answer.

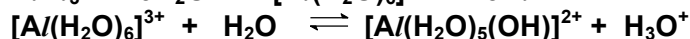
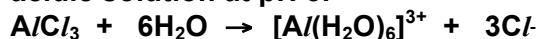
A is magnesium.

There is a large dip in the ionic radius from element C to D, implying that ion formed from element D has one electron shell less. Ion formed from D is P^{3-} which has 18 electrons while ions formed from elements A, B and C have 10 electrons.

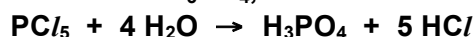
- (ii) Period 3 elements react with chlorine to form chlorides. Account for the difference in the pH values when chloride of **B** and chloride of **D** is separately dissolved in water. Write equations to support your answer.

B is Al while D is P.

Since Al^{3+} has a high charge density, it hydrolyses in water to form an acidic solution at pH 3.



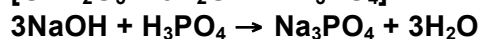
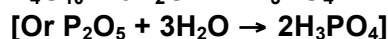
Since P has vacant d orbitals to accept lone pair of electrons from water, it undergoes complete hydrolysis in water completely to form strong acid of HCl and H_3PO_4 , thus the solution formed is at pH 2.



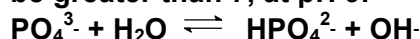
[6]

(b) Period 3 elements also form oxides upon heating with oxygen.

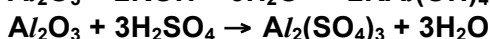
- (i) Write equations to illustrate the reaction between sodium oxide and water as well as phosphorous pentoxide and water. Hence, deduce the pH of the resultant solution after mixing. Support your answers with relevant equations.



PO_4^{3-} from Na_3PO_4 , a strong conjugate base of HPO_4^{2-} , undergoes hydrolysis to give excess OH^- ions, making the pH of the final solution to be greater than 7, at pH 9.



- (ii) Write equations to illustrate the amphoteric behaviour of aluminium oxide, using potassium hydroxide and sulfuric acid in your answer.



[4]

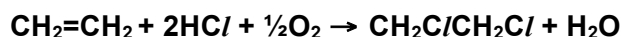
[Total: 10]

- 4 (a) Organic halogen compounds have a wide variety of commercial uses. For example, the polymer, polyvinyl chloride (PVC) is commonly used as insulation of electrical wiring.

The monomer of PVC, $\text{CH}_2=\text{CHCl}$, is industrially produced from $\text{CH}_2=\text{CH}_2$ via the following route:



- (i) Step I is carried out by reacting $\text{CH}_2=\text{CH}_2$ with hydrogen chloride and oxygen over a copper(II) chloride catalyst. Write a balanced equation for this reaction.

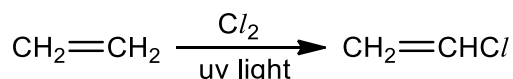


- (ii) Suggest the reagents and conditions that can be used to carry out step II in the laboratory.

Alcoholic NaOH, reflux

[2]

- (b) (i) $\text{CH}_2=\text{CHCl}$ can be formed directly in a single step from $\text{CH}_2=\text{CH}_2$ in the following way:



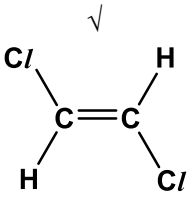
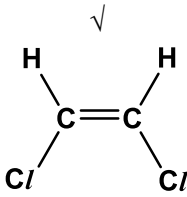
This reaction also results in the formation of a product, $\text{C}_2\text{H}_2\text{Cl}_2$, which exists as a pair of geometric isomers **R** and **S**. Give the name of this product.

1, 2-dichloroethene

- (ii) Some physical properties of the geometric isomers, **R** and **S** are shown below:

Isomer	Melting point / °C	Boiling point / °C
R	−49.4	48.5
S	−81.5	60.2

Draw the structure of **R** and **S** in the boxes below.

R	S
 (<i>trans isomer</i>)	 (<i>cis isomer</i>)

- (iii) $\text{CH}_3\text{CH}_2\text{CH}_3$ can also react with Cl_2 in the presence of uv light to form 2 monochlorinated products, $\text{CH}_2\text{Cl}/\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CHCl}/\text{CH}_3$. It has been determined experimentally that the hydrogen atoms in alkanes are replaced by chlorine atoms at different rates as shown below:

Type of H atom	Reaction	Relative rate of replacement
Primary	$\text{RCH}_3 \rightarrow \text{RCH}_2\text{Cl}$	1
Secondary	$\text{R}_2\text{CH}_2 \rightarrow \text{R}_2\text{CHCl}$	3.5
Tertiary	$\text{R}_3\text{CH} \rightarrow \text{R}_3\text{CCl}$	5

–R represents an alkyl group.

By considering both the number and the rate of replacement of each type of hydrogen atom present in $\text{CH}_3\text{CH}_2\text{CH}_3$, predict the ratio of $\text{CH}_2\text{Cl}/\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CHCl}/\text{CH}_3$ formed. Show your working clearly.

Any one of six H atoms can be replaced to form $\text{CH}_2\text{Cl}/\text{CH}_2\text{CH}_3$ while any one of two H atoms can be replaced to form $\text{CH}_3\text{CHCl}/\text{CH}_3$.

$$\begin{aligned}\text{Hence, the ratio of } \text{CH}_2\text{Cl}/\text{CH}_2\text{CH}_3 \text{ to } \text{CH}_3\text{CHCl}/\text{CH}_3 &= \frac{6}{2} \div 3.5 \\ &= 0.857\end{aligned}$$

[4]

- (c) Chlorofluorocarbons (CFCs) are another class of organic halogen compounds. They were formerly used in aerosol propellants but due to concerns about their effect on the depletion of the ozone layer, they have been replaced by volatile hydrocarbons such as $\text{CH}_3\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$.

Two common CFCs once used in aerosol propellants were CCl_3F and CH_3CClF_2 .

- (i) By comparing the strength of the relevant carbon-halogen bonds, suggest and explain which compound, CCl_3F or CH_3CClF_2 , is likely to have a smaller effect on ozone depletion.

Since the C–Cl bond is weaker than C–F bond, only C–Cl is broken by uv light in the stratosphere to form chlorine radicals which react with the ozone layer.

Thus, since CH_3CClF_2 has a lower chlorine content, it has a smaller effect on ozone depletion.

- (ii) Other than aerosol propellants, give one use of CFCs.

CFCs are used in refrigerants (OR solvents for cleaning electronic circuit boards OR fire extinguishers).

- (iii) State one possible hazard of using hydrocarbons instead of CFCs in aerosol propellants.

Hydrocarbons are highly flammable.

[4]

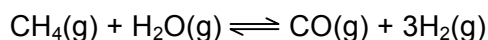
[Total: 10]

Section B

Answer **two** questions from this section on separate answer papers.

- 5 Hydrogen has large scale use in the chemical industry. Although hydrogen is the most abundant element in the Universe, very small quantities of molecular hydrogen have been found to occur naturally.

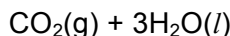
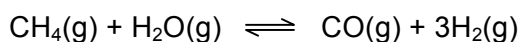
- (a) One of the most efficient methods of producing hydrogen involves steam reforming of natural gas, an equilibrium reaction in which methane gas is reacted with steam to produce carbon monoxide and hydrogen.



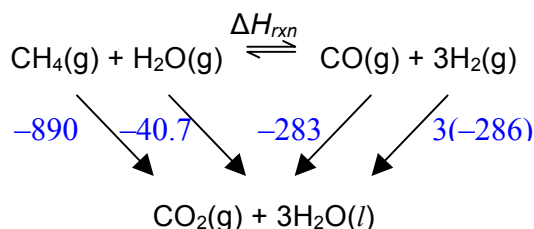
- (i) Use appropriate bond energies given in the *Data Booklet* to calculate an approximate value for the steam reforming of natural gas.
[Take the bond energy for the C≡O bond in CO to be 1077 kJ mol⁻¹]

$$\begin{aligned}\Delta H_{\text{rxn}} &= \text{B.E. (Breaking)} - \text{B.E. (Forming)} \\ &= 4\text{B.E. (C-H)} + 2(\text{O-H}) - 1(\text{C}\equiv\text{O}) - 3(\text{H-H}) \\ &= 4(410) + 2(460) - 1(1077) - 3(436) \\ &= +175 \text{ kJ mol}^{-1}\end{aligned}$$

- (ii) Use the energy cycle and data given below to calculate another approximate value for the steam reforming of natural gas.



$$\begin{aligned}\Delta H_c^\ominus \text{ of CH}_4(\text{g}) &= -890 \text{ kJ mol}^{-1} \\ \Delta H_{\text{vap}} \text{ of H}_2\text{O}(\text{l}) &= +40.7 \text{ kJ mol}^{-1} \\ \Delta H_c^\ominus \text{ of CO}(\text{g}) &= -283 \text{ kJ mol}^{-1} \\ \Delta H_f^\ominus \text{ of H}_2\text{O}(\text{l}) &= -286 \text{ kJ mol}^{-1}\end{aligned}$$



$$\begin{aligned}\Delta H_{\text{rxn}} &= (-890) + (-40.7) - (-283) - 3(-286) \\ &= +210 \text{ kJ mol}^{-1}\end{aligned}$$

- (iii) Suggest two reasons for the discrepancy between the values in (i) and (ii).

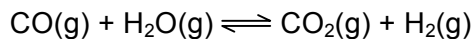
The bond energy values in the *Data Booklet* are average values.

The energy cycle did not account for the energy required to cool the water from 373 K to 298 K OR

The energy cycle did not account for the energy required to heat the water from 298 K to 373 K.

[6]

- (b) The water–gas shift reaction is often used together with the steam reforming of natural gas to produce more hydrogen. This involves the highly exothermic reaction of carbon monoxide with steam to produce carbon dioxide and hydrogen.



The equilibrium constant, K_c is found to be 0.64 at 1000 K.

- (i) Write an expression for K_c for this reaction.

$$K_c = \frac{[\text{CO}_2\text{(g)}] \cdot [\text{H}_2\text{(g)}]}{[\text{CO(g)}] \cdot [\text{H}_2\text{O(g)}]}$$

- (ii) A mixture containing 0.80 mol of CO, 0.80 mol of H₂O, 0.40 mol of CO₂ and 0.40 mol of H₂ was placed in a 2 dm³ flask and allowed to come to equilibrium at 1000 K.

Calculate the amount, in mol, of each substance present in the equilibrium mixture at 1000 K.

Let the change in concentration of CO(g) be x.

	CO(g)	H ₂ O(g)	$\rightleftharpoons$	CO ₂ (g)	H ₂ (g)
Initial conc./ mol	0.80/2	0.80/2		0.40/2	0.40/2
Change in conc./ mol	-x	-x		+x	+x
Eqm. conc./ mol dm ⁻³	0.40-x	0.40-x		0.20+x	0.20+x

$$K_c = \frac{[\text{CO}_2\text{(g)}] \cdot [\text{H}_2\text{(g)}]}{[\text{CO(g)}] \cdot [\text{H}_2\text{O(g)}]}$$

$$\Rightarrow 0.64 = \frac{[0.20 + x]^2}{[0.40 - x]^2}$$

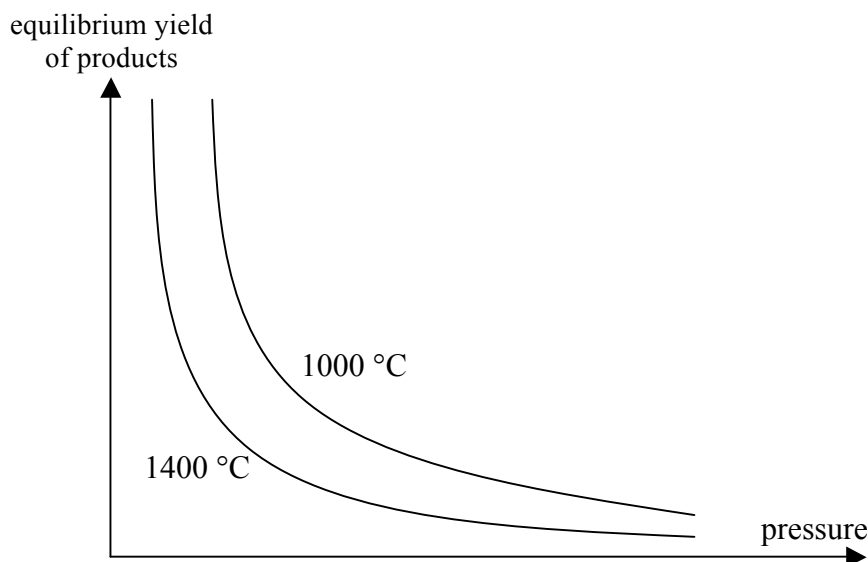
$$\Rightarrow x = 0.0667 \text{ mol dm}^{-3}$$

$$n_{\text{CO(g)}} = n_{\text{H}_2\text{O(g)}} = 2(0.40 - 0.0667) = 0.667 \text{ mol}$$

$$n_{\text{CO}_2\text{(g)}} = n_{\text{H}_2\text{(g)}} = 2(0.20 + 0.0667) = 0.534 \text{ mol}$$

[5]

- (c) The graph below shows the variation in equilibrium yield of the products with temperature and pressure for the water–gas shift reaction.



- (i) State *Le Chatelier's Principle*.

***Le Chatelier's Principle* states that if a system at equilibrium is subjected to a change which disturbs the equilibrium, the position of equilibrium will shift in a direction to reduce the change.**

- (ii) Hence comment on the accuracy of the graph with respect to both temperature and pressure.

***Le Chatelier's Principle* states that if temperature increases, equilibrium position will shift to favour the endothermic side to absorb the excess heat. Since the forward reaction is exothermic, equilibrium position will shift to the left, resulting in a decrease in equilibrium yield. Hence the graph is correct with respect to temperature.**

***Le Chatelier's Principle* states that if pressure increases, equilibrium position will shift to side with less gaseous particles to reduce the pressure. However, since the no. of moles of gaseous reactants is equal to the no. of moles of gaseous products, equilibrium yield remain constant with increasing pressure. Hence the graph is incorrect with respect to pressure.**

[5]

- (d) In the presence of a catalyst mixture made up of alumina, copper and zinc oxide, carbon monoxide and hydrogen reacts to form compound **D** instead.

Compound **D** is commonly obtained from biomass and consist only of the elements C, H and O. When 0.55 g of **D** was completely burnt in air, the heat produced raised the temperature of 100 g of water by 29.4 °C. The standard enthalpy change of combustion of **D** is found to be -715 kJ mol^{-1} .

- (i) Calculate the amount of heat released in the experiment.
[Assume the heat capacity of all solutions to be $4.18 \text{ J K}^{-1} \text{ cm}^{-3}$]

$$\begin{aligned} \text{Heat released} &= m \times c \times \Delta T \\ &= 100 \times 4.18 \times 29.4 \\ &= 12300 \text{ J [OR 12.3 kJ]} \end{aligned}$$

- (ii) Hence deduce the molecular formula of compound **D**.

$$\Delta H_c = - \frac{\text{heat released}}{n_D} \Rightarrow -715 \times 1000 = - \frac{12300}{n_D}$$

$$n_D = - \frac{12300}{715000} = 0.0172 \text{ mol}$$

$$\text{Molar mass of D} = \frac{m}{n_D} = \frac{0.55}{0.0172} = 32.0$$

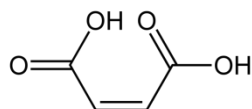
Molecular formula of D = CH₃OH [OR CH₄O]

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[Total: 20]

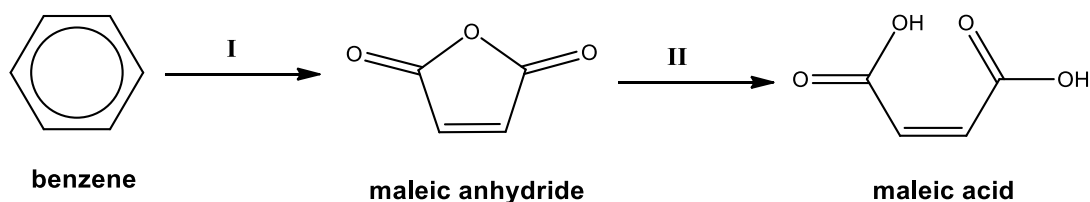
- 6 "The bubble has burst for several tea businesses here" quoted from The Straits Times on 28 May 2013.

The Agri-Food and Veterinary Authority of Singapore (AVA) recalled 11 types of tapioca starch balls - also known as "pearls" - made in Taiwan that are used in popular 'bubble teas', after they were found to contain maleic acid. Consuming high levels of maleic acid can cause kidney damage.



Maleic acid

- (a) Maleic acid is commonly produced via the following 2-step synthetic route industrially.

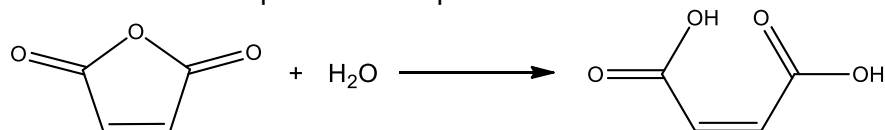


- (i) State the type of reactions for **I** and **II**.

I – oxidation

II – hydrolysis

- (ii) Write a balanced equation for step **II**.



- (iii) Suggest how you can distinguish the following pairs of compounds:

I – Benzene and maleic anhydride

II – Maleic anhydride and maleic acid

In each case, state the reagents and conditions as well as observations for each compound.

I – Benzene and maleic anhydride

Add aqueous bromine (or Br_2 in CCl_4) at room temperature.

Reddish-brown solution turns colourless for maleic anhydride. No colour change observed for benzene.

II – Maleic anhydride and maleic acid

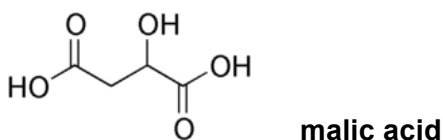
Add solid sodium carbonate (or $\text{NaHCO}_3(\text{s})$) at room temperature.

Effervescence which produced a white ppt in limewater observed for maleic acid. No effervescence observed for maleic anhydride.

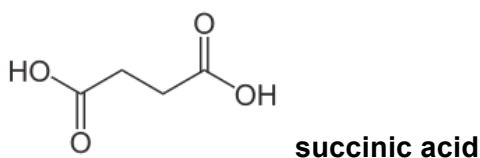
[7]

(b) Draw the structure of the major product formed when maleic acid reacts with

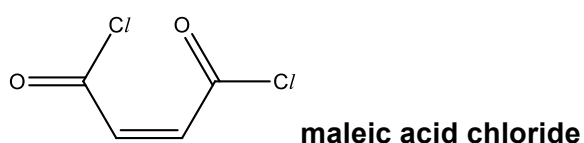
- (i) concentrated sulfuric acid, followed by hydrolysis



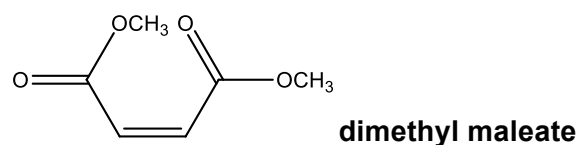
- (ii) hydrogen gas with palladium on carbon



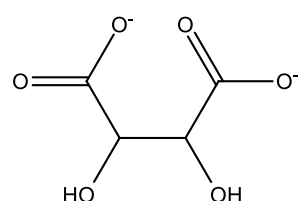
- (iii) thionyl chloride



- (iv) methanol with concentrated sulfuric acid

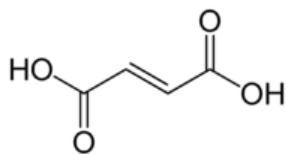


- (v) cold alkaline potassium manganate(VII)



[6]

(c) (i) State the type of isomerism displayed by maleic acid and draw its isomer, fumaric acid.

Geometric (or cis-trans) isomerism.**fumaric acid**

- (ii) Based on the structures of maleic acid and fumaric acid, predict which acid has a lower melting point.

Maleic acid has a lower melting point.

Intramolecular hydrogen bonding can take place in maleic acid due to the close proximity of the –OH and =O groups. As such, only weaker dipole-dipole interactions (or less extensive intermolecular hydrogen bonding) are present between maleic acid molecules unlike the stronger (or more extensive) intermolecular hydrogen bonding between fumaric acid molecules. Less energy is required to overcome the weaker intermolecular interactions, which results in the lower melting point.

- (iii) A student added some water to a mixture of maleic acid and fumaric acid and performed a simple filtration. He concluded that the residue obtained must be fumaric acid.

Do you agree with the student's statement? Justify your answer.

Yes, I agree.

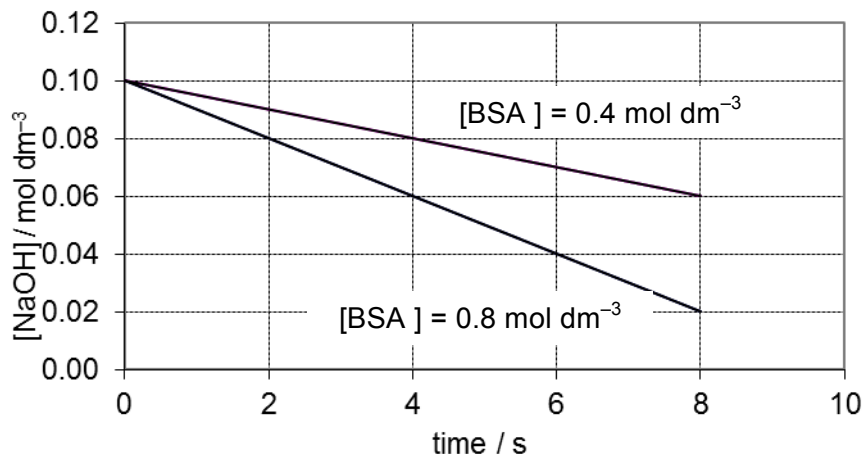
Fumaric acid is less soluble than maleic acid, since more energy is required to overcome the stronger solute-solute interaction, (or more extensive) hydrogen bonding in fumaric acid, as compared to the weaker dipole-dipole interactions (or less extensive intermolecular hydrogen bonding) in maleic acid.

Note: Both fumaric acid and maleic acid form strong hydrogen bonds with water (solute-solvent interactions). Strong hydrogen bonding exists between water molecules too (solvent-solvent interactions).

[7]

[Total: 20]

- 7 (a) Benzenesulfonic acid (BSA) reacts with aqueous NaOH to produce phenol. The graph below shows the reaction between aqueous sodium hydroxide and solutions of benzenesulfonic acid of different concentration at temperature $T\text{ }^{\circ}\text{C}$.



- (i) With reference to the graph above, deduce the order of reaction with respect to sodium hydroxide and benzenesulfonic acid and hence write down the rate equation.

Gradient of graph with $[\text{BSA}] = 0.4\text{ mol dm}^{-3}$ is half that of graph with $[\text{BSA}] = 0.8\text{ mol dm}^{-3}$.

Hence, the reaction is first order with respect to BSA.

Both graphs are straight lines with a constant gradient, which implies that change in concentration of NaOH has no impact on the rate. Hence, the reaction is zero order with respect to NaOH.

Rate = $k[\text{BSA}]$

- (ii) Suggest a method to monitor the concentration of NaOH in the above kinetic study.

Use a pH probe to monitor the pH of the mixture. Or titrate NaOH with a standard solution of HCl at regular time intervals.

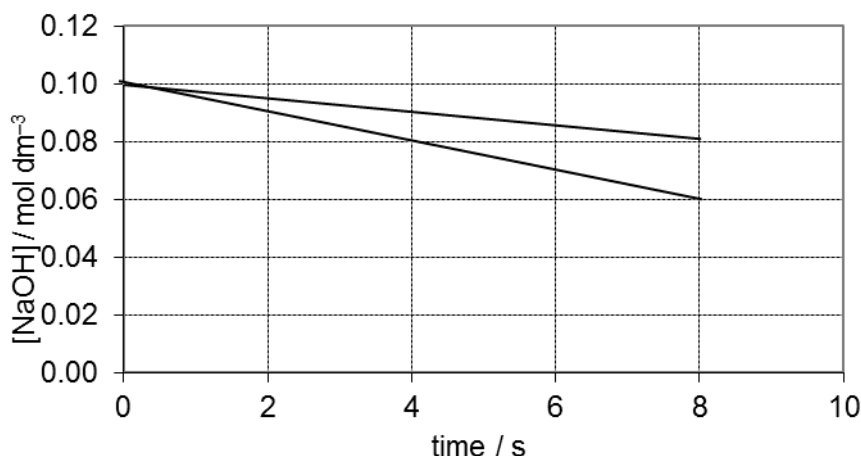
- (iii) Determine the time taken for the concentration of benzenesulfonic acid to reduce from 0.4 mol dm^{-3} to half of its original concentration.

Rate = $k[\text{BSA}] = 0.04/8 = 0.0005\text{ mol dm}^{-3}\text{ s}^{-1}$.

Therefore, $k = 0.0005/0.4 = 0.0125$

$t_{1/2} = \ln 2/k = 55.5\text{ s}$

- (iv) On the grid drawn, sketch and label a graph for the reaction of 0.10 mol dm^{-3} of aqueous sodium hydroxide with 0.4 mol dm^{-3} of BSA at $(T-10)\text{ }^{\circ}\text{C}$.

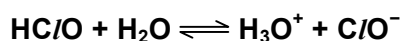


- (v) Account for the shape of the graph in part (iv).

For every 10 °C decrease in temperature, the rate is decreased by 2 times. •As the order of the reaction with respect to the concentration of NaOH does not change, the gradient of the graph at (T–10) °C is halved of that for T °C.

- (b) Hypochlorous acid, HC/O, is a *weak acid* that is formed when chlorine reacts with water.

- (i) With the aid of an equation, explain what is meant by weak acid.



The acid undergoes partial dissociation to produce H⁺.

- (ii) Write an expression for the acid dissociation constant, K_a , for HC/O.

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{C/O}^-]}{[\text{HC/O}]}$$

- (iii) The following equilibrium concentrations were determined for a solution of HC/O, calculate the value of K_a of HC/O, stating the units.

Species	Concentration / mol dm ⁻³
HC/O(aq)	0.0200
H ⁺ (aq)	2.50×10^{-5}
C/O ⁻ (aq)	2.50×10^{-5}

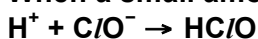
$$K_a = \frac{(2.50 \times 10^{-5})^2}{0.02} = 3.13 \times 10^{-8} \text{ mol dm}^{-3}$$

[3]

- (c) A solution containing hypochlorous acid, HC/O, and sodium chlorate(I), NaC/O, can act as a buffer solution. Explain, with the help of equations, how this solution can regulate pH when a relatively small amount of acid or base is added to the solution.

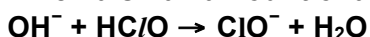
The solution contains a large reservoir of HC/O and C/O⁻.

When a small amount of acid is added, it will be neutralised by C/O⁻ as shown:



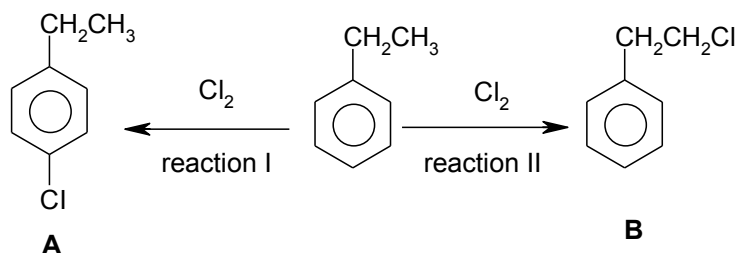
A large reservoir of C/O⁻ ensures sufficient removal of H⁺ to maintain pH.

When a small amount of alkali is added, it will be neutralised by HC/O:



A large reservoir of HC/O ensures sufficient removal of OH⁻ to maintain pH.

- (d) Ethylbenzene can react with chlorine in two ways, depending on the conditions of the reaction.



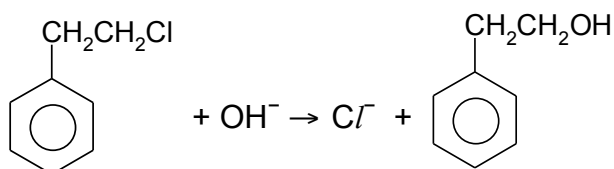
- (i) State the condition needed for reaction I.

Anhydrous FeCl_3 or AlCl_3

- (ii) Describe a chemical test to distinguish between **A** and **B**. State the reagents, conditions, and the observations for each compound. Write balanced equations for any reactions that occur.

Boil each compound with NaOH(aq) , cool and acidify with $\text{HNO}_3\text{(aq)}$ followed by adding $\text{AgNO}_3\text{(aq)}$.

B forms a white ppt, whereas no ppt is observed for A.



$\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl}$ (white ppt)